

Homework 4

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We study two realizations of the action of the symmetric group S_n for finite n :

- **Index action:** S_n acts on $X = \{1, 2, \dots, n\}$ by $g \cdot i = g(i)$.
- **Coordinate action:** S_n acts on $v \in \mathbb{R}^n$ by permuting coordinates: $(gv)_i = v_{g^{-1}(i)}$.

These actions are equivalent, as the coordinate action is the index action applied to the vector indices. The standard permutation representation is $\rho : S_n \rightarrow GL(n, \mathbb{R})$, with $\rho(g)_{i,j} = \delta_{j,g(i)}$.

Let $G \leq S_n$. Let's look at the G -equivariant linear maps $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$, such that

$$T\rho(g)v = \rho(g)Tv \quad (1)$$

for all $g \in G, v \in \mathbb{R}^n$.

We can solve (1) in two ways:

- **Orbits of Indexes:** Looking on the condition (1) applied to the indexes of T .
- **Character Theory:** Looking on the condition (1) applied to the representation ρ of G .

Orbits of Indexes Since G acts on the indexes of T , we have

$$\rho(g)T\rho(g)^{-1} = T, \quad (2)$$

for all $g \in G$ lead to the condition

$$T_{i,j} = T_{g^{-1}(i),g^{-1}(j)}. \quad (3)$$

This holds for all $g \in G$ and $i, j \in X$, therefore, for each pair of indexes (i, j) , the value $T_{i,j}$ is constant on the orbit $Orb_G((i, j))$. Leading to the conclusion that the number of free parameters in T (the dimensionality $\dim HOM_G(\mathbb{R}^n, \mathbb{R}^n)$) is equal to the number of orbits of the action of G on X^2 . The number of orbits can be computed using Burnside's lemma as in Tutorial 12 Problem 2

$$|X^2/G| = \frac{1}{|G|} \sum_{g \in G} |X^g|^2 \quad (4)$$

where $X^g = \{x \in X \mid g \cdot x = x\}$ is the set of fixed elements of g .

Character Theory In the course we have seen the proof that the dimensionality of the space of G -linear transformations $HOM_G(\mathbb{R}^n, \mathbb{R}^n)$ is equal to

$$\dim HOM_G(\mathbb{R}^n, \mathbb{R}^n) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(\rho(g))^2. \quad (5)$$

From condition (2) we have

$$(\rho(g) \otimes \rho(g)) \text{vec}(T) = \text{vec}(T). \quad (6)$$

We can see that $\text{vec}(T)$ is an eigenvector of $\rho(g) \otimes \rho(g)$ with eigenvalue 1 for all $g \in G$. Moreover, let $\{O_1, \dots, O_k\}$ be the orbits of the action of G on X^2 , where k is the number of orbits; and let $\mathbf{1}_{O_i} \in \mathbb{R}^{n \times n}$ be the indicator matrix of the orbit O_i , i.e. $\mathbf{1}_{O_i}(i, j) = 1$ if $(i, j) \in O_i$ and 0 otherwise. Then, $\text{vec}(\mathbf{1}_{O_i})$ must be an eigenvector of $\rho(g) \otimes \rho(g)$ with eigenvalue 1 for all $g \in G$ as well, since it is constant when g acts on it. Define $U^G = \{v \in \mathbb{R}^{n \times n} \mid (\rho(g) \otimes \rho(g))v = v, \text{ for all } g \in G\}$, as the fixed subspace of $\mathbb{R}^{n \times n}$ under the action of G . U^G is isomorphic to the space of G -linear transformations $HOM_G(\mathbb{R}^n, \mathbb{R}^n)$. In the course we saw that the dimension of U^G is equal $\text{tr}(\pi_{U^G})$, where π_{U^G} is the projection operator onto U^G . We showed that $\pi_{U^G} = \frac{1}{|G|} \sum_{g \in G} \rho(g) \otimes \rho(g)$, and therefore we have

$$\dim HOM_G(\mathbb{R}^n, \mathbb{R}^n) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(\rho(g) \otimes \rho(g)) = \frac{1}{|G|} \sum_{g \in G} \text{tr}(\rho(g))^2. \quad (7)$$

This is the same as (5), and each eigenvalue of $\rho(g) \otimes \rho(g)$ corresponds to an orbit of the action of G on X^2 . It is clear that each orbit corresponds to a unique eigenvector of $\rho(g) \otimes \rho(g)$, but the other way need to be explained. If we have an eigenvector v of $\rho(g) \otimes \rho(g)$, it must satisfy (3), which means that it must be constant on the orbits, therefore it can be represented as a linear combination of the $\{\text{vec}(O_i)\}_{i=1}^k$, and therefore it cannot add a dimension to the eigenvalues corresponding to the orbits. In the case of index action X^g corresponding to the fixed points of g acting on the indexes. In the case of coordinate action, U^G corresponds to the fixed vector space of G acting on the vectorization of the matrices T .

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2.1

The natural group here is the cross product of the permutation groups $S_n \times S_m$ acts on the Netflix rating matrices $A \in \mathbb{R}^{n \times m}$. It can be explained by the fact that the order of the m movies and the order of the n users is arbitrary and both can be changed independently.

2.2

Since we look at the G -linear transformation from matrices in $\mathbb{R}^{n \times m}$ to itself, it will be more convenient to consider the vectorization of these matrices. Thus $\text{vec}(A) = [a_{1,1}, \dots, a_{n,1}, \dots, a_{1,m}, \dots, a_{n,m}]^T \in \mathbb{R}^{nm}$. Let $g = (g_l, g_r) \in S_n \times S_m$ be a group element where $g_l \in S_n$ is a left action that acts on the rows of A and $g_r \in S_m$ acts on the columns of A as follows:

$$gA = g_l A g_r^T = P_l A P_r^T, \quad (8)$$

where $P_l \in \mathbb{R}^{n \times n}$ and $P_r \in \mathbb{R}^{m \times m}$ are the permutations of g_l and g_r , respectively. Define $v = \text{vec}(A) \in \mathbb{R}^{nm}$. We are actually looking on the isomorphic group $G \leq S_{nm}$, which acts on the indexes of the vectorization of A . We will use G and $S_n \times S_m$ interchangeably. In index notation we have $(gA)_{i,j} = A_{g_l^{-1}(i), g_r^{-1}(j)}$. We are looking at the affine transformation that is equivariant, thus $f(v) = Wv + b$, that fulfills for all $g \in S_n \times S_m$ and $v \in \mathbb{R}^{nm}$

$$\begin{aligned} f(gv) &= W(gv) + b = g(Wv + b) \\ &\Rightarrow Wgv + b = gWv + gb \end{aligned} \quad (9)$$

Since (9) needs to be fulfilled for all $v \in \mathbb{R}^{nm}$ it leads to $Wg = gW$ and equivalently $gWg^T = W$. Moreover, it requires that $gb = b$ for all $g \in G$. We have

$$gA = g_l A g_r^T = P_l A P_r^T. \quad (10)$$

By vectorizing the above we obtain

$$\text{vec}(P_l A P_r^T) = (P_r \otimes P_l) \text{vec}(A) \Rightarrow P_{nm} = (P_r \otimes P_l) \quad (11)$$

For b , we can see from (9) that this means that b is constant, $b = c \cdot \mathbf{1}$. This can be seen by applying the inverse vectorization operation on b . Since the condition (9) requires that b will be invariant to the action of g , and by acting on b we can replace each index $\text{vec}^{-1}(b_{i,j})$ with any other index. Therefore, all values of b required to be the same.

Let us look at the action of the permutation $g^{-1} = (g_l^{-1}, g_r^{-1})$ on the vector v . We take the inverse g^{-1} for convenience of notation. Originally, the element with the index $A_{i,j}$ moves to $v_{i+n(j-1)}$, therefore

$$(g^{-1}A)_{i,j} = A_{g_l(i), g_r(j)} = v_{g_l(i) + n(g_r(j)-1)}. \quad (12)$$

Consequently, for an index $1 \leq k \leq nm$, define $j = k/n$ as the truncating integer division and $i = k \% n$ as the modulo operation, thus $(g^{-1}v)_k = v_{g_l(i) + n(g_r(j)-1)}$. Let k, l be indexes in $[1, \dots, nm]$ with the decomposition $k = i_1 + n(j_1 - 1)$ and $l = i_2 + n(j_2 - 1)$. Then $(g^{-1}Wg)_{k,l} = W_{g_l(i_1) + n(g_r(j_1)-1), g_l(i_2) + n(g_r(j_2)-1)}$.

2.2.1 Orbit analysis

Let us look at the possible orbits in the index set $X^2 = \{1, 2, \dots, nm\}^2$. Indexes (k', l') can be on the same orbits as (k, l) if and only if there exists a permutation $(g_l, g_r) \in S_n \times S_m$ such that

$$\begin{cases} k' &= g_l(i_1) + n(g_r(j_1) - 1) \\ l' &= g_l(i_2) + n(g_r(j_2) - 1) \end{cases} \quad (13)$$

Define i_3, i_4, j_3, j_4 such that $k' = i_3 + n(j_3 - 1)$ and $l' = i_4 + n(j_4 - 1)$. It is clear that we can create equivalence groups by the rule:

$$(i_1 \neq i_2) \wedge (j_1 \neq j_2). \quad (14)$$

It is clear that $i_3 = i_4$ holds if and only if $i_1 = i_2$, since by applying g_l we have $g_l(i_1) = g_l(i_2) = i_3 = i_4$, and since g_l is a bijection. The same goes for $j_3 = j_4$.

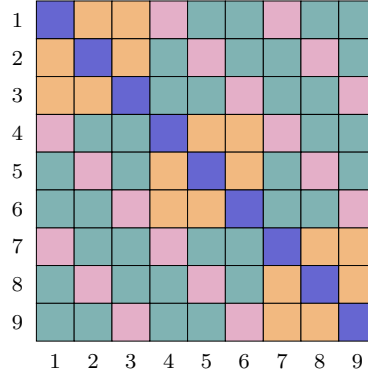
These are the orbits and the explanation for why they are the only orbits is as follows: if $i_1 = i_2$ then we can move to any other k', l' that has $i_3 = i_4$ by applying the right $g_l(i_1) = i_3$. If $i_1 \neq i_2$

we can move to any other indexes $i_3 \neq i_4$ since S_n is 2-transitive. The same goes for the indexes j . Therefore, we will have only these 4 orbits:

$$\begin{aligned} O_{=i=j} &: i_1 = i_2, j_1 = j_2 \\ O_{=i \neq j} &: i_1 = i_2, j_1 \neq j_2 \\ O_{\neq i=j} &: i_1 \neq i_2, j_1 = j_2 \\ O_{\neq i \neq j} &: i_1 \neq i_2, j_1 \neq j_2 \end{aligned}$$

2.3

For $n = m = 3$ we will obtain $W \in \mathbb{R}^{9 \times 9}$ and $b \in \mathbb{R}^9 = c \cdot \mathbf{1}$



2.4

In the case of a linear invariant layer, for all $v \in \mathbb{R}^{nm}$ and $g \in G$ we have:

$$\begin{aligned} f(gv) &= f(v) \\ \Rightarrow w^T gv + b &= w^T v + b \\ \Rightarrow g^T w &= w \end{aligned}$$

For $w \in \mathbb{R}^{nm}$, this condition is equivalent to the condition on b in the previous clause. Therefore, $w = c_2 \cdot \mathbf{1}$.

2.5

In the case of 3-tensors with the symmetric group is $S_{N_1} \times S_{N_2} \times S_{N_3}$ applied independently to each index axis of the tensor. We have:

$$v = \text{vec}(A) \quad v_k = a_{i_1, i_2, i_3}, \quad (15)$$

where $k = i_1 + N_1(i_2 - 1) + N_1N_2(i_3 - 1)$. Define $k = i_1^k + N_1(i_2^k - 1) + N_1N_2(i_3^k - 1)$ and $l = i_1^l + N_1(i_2^l - 1) + N_1N_2(i_3^l - 1)$. We will have the same derivation as in the previous clause 2.2, thus

$$W_{k,l} = W_{g(k),g(l)} \quad (16)$$

The indexes (k, l) can transform to

$$k' = g_1(i_1^k) + N_1(g_2(i_2^k) - 1) + N_1N_2(g_3(i_3^k) - 1) \quad (17)$$

$$l' = g_1(i_1^l) + N_1(g_2(i_2^l) - 1) + N_1N_2(g_3(i_3^l) - 1) \quad (18)$$

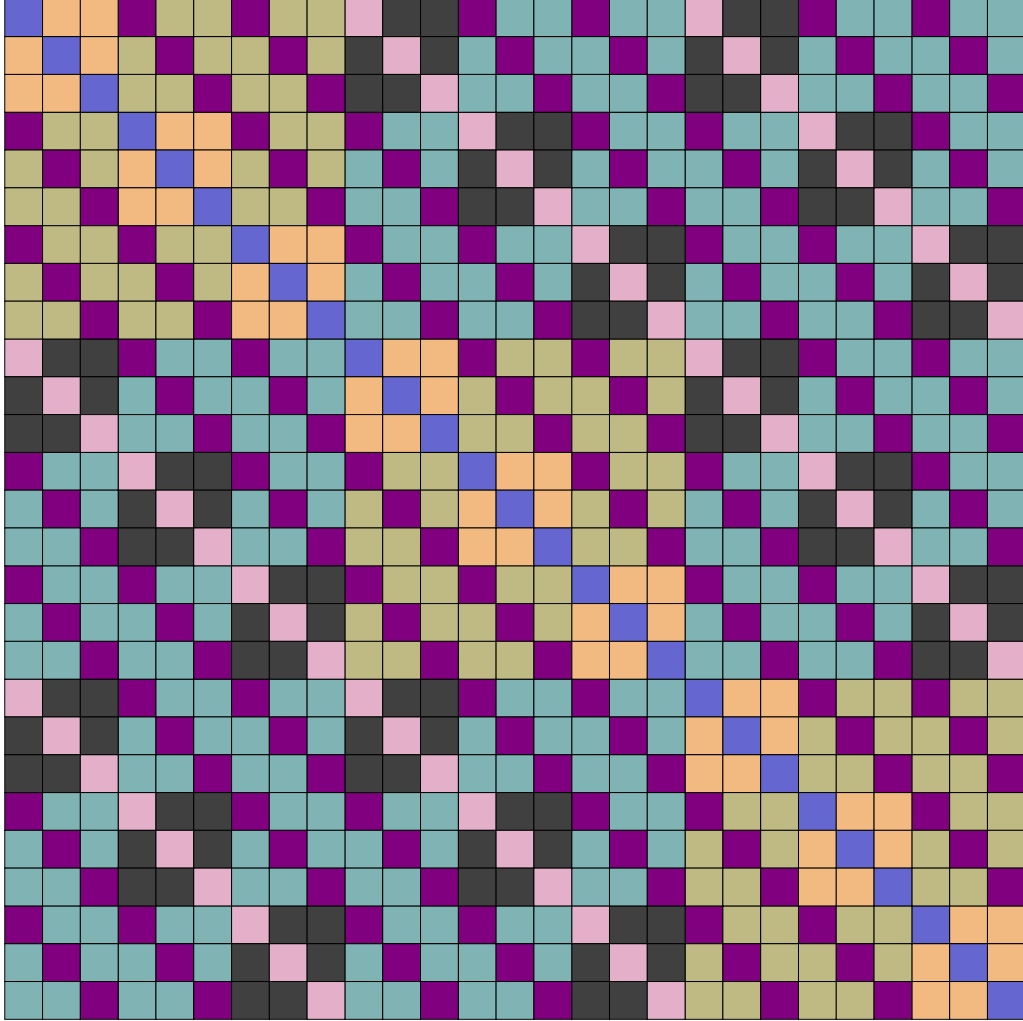
where the equivalence relations follow the same pattern as before, thus the orbits are

$$\bigwedge_{n=1}^3 i_n^k \neq i_n^l. \quad (19)$$

As a result, the orbits are:

$$\begin{aligned} O_{=i_1=i_2=i_3} &: i_1^k = i_1^l, i_2^k = i_2^l, i_3^k = i_3^l \\ O_{=i_1=i_2 \neq i_3} &: i_1^k = i_1^l, i_2^k = i_2^l, i_3^k \neq i_3^l \\ O_{=i_1 \neq i_2=i_3} &: i_1^k \neq i_1^l, i_2^k = i_2^l, i_3^k = i_3^l \\ &\vdots \\ O_{\neq i_1 \neq i_2 \neq i_3} &: i_1^k \neq i_1^l, i_2^k \neq i_2^l, i_3^k \neq i_3^l \end{aligned}$$

For $N_1 = N_2 = N_3 = 3$, we obtain $W \in \mathbb{R}^{27 \times 27}$ and $b \in \mathbb{R}^{27}$.

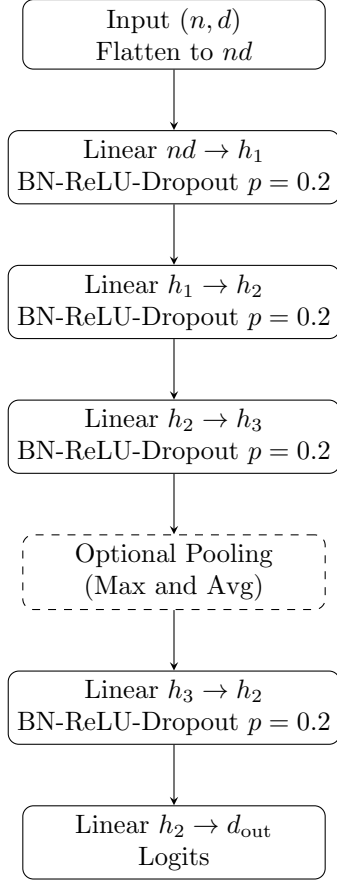


Question 4 - 3D Point Cloud Classification

Description of the models

We have two architectures for the 3D point cloud classification. One is named BasePointCloudNet which, as the name suggests, is a base architecture for the point cloud classification which the following network use: canonization-based network, symmetrization based network, sample symmetrization based network, and augmentation-based network. The second architecture is called equivariant-based network, which is an invariance network with equivariant layers. The equivariant-based network is a more advanced architecture that account for permutation invariance of the point cloud data, intrinsically. The architecture of BasePointCloudNet is given in Figure 1.

BasePointCloudNet MLP architecture



Typical sizes: $nd = 256 \cdot 3$, $h_1 = 128$, $h_2 = 256$, $h_3 = 256$, $d_{\text{out}} = 40$.

Figure 1: MLP used in BasePointCloudNet: flattened point cloud feeds a 5-layer MLP with BN, ReLU, Dropout, and an optional pooling layer between the 3rd and 4th layers.

The second architecture, the equivariant-based network, is the same as the BasePointCloudNet, but the first 3 layers are equivariant. Then max-pooling and mean-pooling are applied to the output of the equivariant layers, and the results are concatenated and fed to invariant two layers. In the equivariant-based network, the hidden layers are given by $h_1 = 64$, $h_2 = 128$, $h_3 = 256$. The number of parameters is 254248. In both architectures we used cross-entropy loss function.

For the architectures: symmetrization, sampled-symetrization, and augmentation, we used

We used a computer with RTX 2080 Ti GPU, with 12GB of memory and everything ran on it. The models run for 1000 epochs with the condition that the accuracy on the validation set will be more than 80%, which no model achieved and therefore all the models under all the settings

are considered to run out of time. The model was taken according to the best accuracy on the validation set. The data was split to 80% training, 10% validation, and 10% test. From the test set a subset was taken according to the requirements in the question.

Accuracy

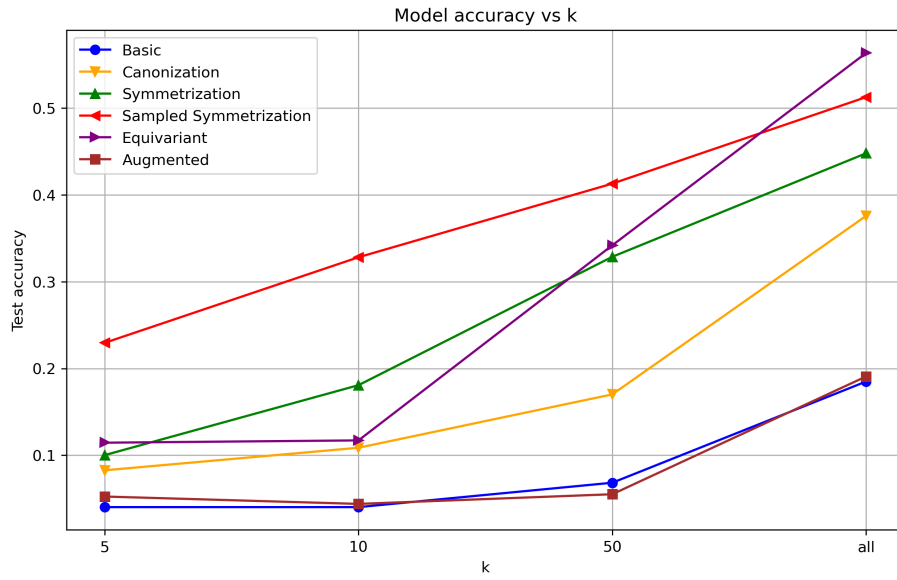


Figure 2: Accuracy of the models on the test set as a function of k (the number of training example for each class)

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